

Introduction to Robotics & Intelligent Control Systems

End Semester Examination — Paper II

Full Marks: **100** Time: **3 Hours** Equal weightage: Mechanical & Electrical

Attempt ALL questions. Show complete working. Formula cheat-sheet attached on last page.

Paper Structure: **Sec A** Short Answer (20 marks) · **Sec B** Kinematics & HTM (25 marks) · **Sec C** Jacobian & Statics (15 marks) · **Sec D** DC Motors (20 marks) · **Sec E** Circuits: R, L, C & Control (20 marks)

SECTION A — Short Answer & Conceptual 20 Marks (2 each)

Answer ALL. Each question carries 2 marks.

QA1. A rotation matrix R satisfies $R^T R = I$ and $\det(R) = +1$. Using these two properties alone, prove that $R^{-1} = R^T$. [2]

Solution QA1

Since $R^T R = I$ by the orthogonality property, and the inverse of a matrix A is defined as the matrix B such that $AB = I$, we directly identify $B = R^T$. Therefore $R^{-1} = R^T$. The condition $\det(R) = +1$ additionally ensures this is a **proper** rotation (no reflection), guaranteeing the inverse is physically meaningful as a rotation in the same group $SO(3)$.

QA2. Explain physically why **rotations in 3D are non-commutative** with a concrete example. [2]

Solution QA2

Hold a book flat. **Rotation 1:** rotate 90° about Z (spine rotates to your left). **Rotation 2:** rotate 90° about X (front cover faces up). The final orientation depends on order:

$$R_z R_x \neq R_x R_z$$

Physically, each rotation changes the body-frame axes, so the second rotation acts on a *different axis configuration* depending on which went first. Matrix multiplication is non-commutative in general, and rotation matrices inherit this.

QA3. State **Kirchhoff's Current Law** and apply it to a node where currents of 3 A, -1 A, and I all meet. Find I . [2]

Solution QA3

KCL: The algebraic sum of all currents entering (or leaving) any node equals zero: $\sum I_k = 0$.

Taking “into node” as positive:

$$3 + (-1) + I = 0 \implies I = -2 \text{ A}$$

The negative sign means 2 A actually *leaves* the node. $I = -2 \text{ A}$

QA4. What is the **back-EMF** in a DC motor and what physical effect does it represent? Why does it limit the motor's maximum speed? [2]

Solution QA4

Back-EMF (counter-electromotive force) $E_b = K_e\omega$ is the voltage generated by the rotating armature coil moving through the magnetic field (Faraday's law). It *opposes* the supply voltage. The motor's armature current is:

$$I = \frac{V - E_b}{R} = \frac{V - K_e\omega}{R}$$

As ω increases, E_b increases, reducing current and torque. At no-load, $\omega \rightarrow V/K_e$ and $I \rightarrow 0$ — this is the maximum (no-load) speed. The motor cannot spin faster under its own supply voltage.

QA5. A capacitor $C = 10\ \mu\text{F}$ has voltage $v_C = 5\ \text{V}$ (constant, DC steady state). What is the current through it and why? [2]

Solution QA5

$$i_C = C \frac{dv_C}{dt} = C \cdot 0 = \boxed{0\ \text{A}}$$

A capacitor is a **DC open circuit** at steady state. Since DC implies $dv/dt = 0$, no current flows through it. This is why capacitors block DC but pass AC signals.

QA6. For a **half-wave rectifier** with peak voltage $V_m = 20\ \text{V}$ across load $R = 10\ \Omega$, find the DC (average) output voltage and average load current. [2]

Solution QA6

A half-wave rectifier conducts for only the positive half-cycle ($0 \leq \omega t \leq \pi$):

$$V_{dc} = \frac{V_m}{\pi} = \frac{20}{\pi} \approx \boxed{6.37\ \text{V}}$$

Average load current:

$$I_{dc} = \frac{V_{dc}}{R} = \frac{6.37}{10} \approx \boxed{0.637\ \text{A}}$$

QA7. In a MOSFET-based H-bridge, what is a **shoot-through fault**? Why is it catastrophic? Name one hardware protection used against it. [2]

Solution QA7

A shoot-through fault occurs when both the **high-side and low-side MOSFETs on the same leg** are turned ON simultaneously, creating a direct short-circuit path from V_{cc} to GND (bypassing the motor). The only impedance is parasitic resistance/inductance, leading to:

$$\left. \frac{di}{dt} \right|_{t=0+} = \frac{V_{cc}}{L_{\text{parasitic}}} \gg 0$$

This causes catastrophic current spike destroying the MOSFETs and power supply.

Protection: *Dead-time insertion* — a brief delay (typically 100 ns–1 μs) is enforced

between turning one MOSFET off and the complementary one on, preventing both from being ON simultaneously.

QA8. What does the **DH parameter** α_i (link twist) represent geometrically? What value does it take for all links of a planar robot, and why? [2]

Solution QA8

α_i is the angle between axes Z_{i-1} and Z_i measured *about* the common normal (X_i axis). It captures how much consecutive joint axes are “twisted” relative to each other.

For a **planar robot**, all joint axes ($Z_0, Z_1, \dots, Z_n$) are parallel and perpendicular to the plane of motion. Since parallel axes have zero angle between them:

$$\alpha_i = 0^\circ \text{ for all links of a planar robot.}$$

QA9. Why does an inductor act as a **short circuit** at DC steady state but as a high-impedance element at high frequency? [2]

Solution QA9

The voltage across an inductor is $v_L = L \frac{di}{dt}$.

DC steady state: Current is constant, $di/dt = 0 \implies v_L = 0$. With zero voltage drop and non-zero current, the inductor behaves as a **short circuit** (like a plain wire).

High frequency AC: Inductive impedance $Z_L = j\omega L$. As frequency $\omega \rightarrow \infty$, $|Z_L| \rightarrow \infty$ — the inductor **blocks** high-frequency currents. This makes inductors useful as low-pass filters in motor drive circuits (smooth ripple current).

QA10. A 3R planar robot is commanded to reach a point strictly *inside* its workspace. How many real solutions does the inverse kinematics problem typically have, and why? [2]

Solution QA10

A 3R planar manipulator reaching a target with specified (x, y) position only (not orientation) has **2 solutions** in general: **elbow-up** and **elbow-down** (like a 2R robot, with the third joint absorbing orientation). If the full pose (position + orientation) is specified, the 3R system is exactly-determined with typically **a unique solution** (or finite number). Points near workspace boundaries may have fewer solutions.

SECTION B — Kinematics & Homogeneous Transformations 25 Marks

QB1. A robot end-effector first rotates 45° about the X -axis, then 30° about the Y -axis. The final position is $(2, 5, 3)^T$.

(a) Compute $R_x(45^\circ)$ and $R_y(30^\circ)$ and find the overall rotation $R = R_y(30^\circ)R_x(45^\circ)$. [5]

(b) Write the complete 4×4 HTM. [2]

[7]

Solution QB1

(a)

$$R_x(45^\circ) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 45^\circ & -\sin 45^\circ \\ 0 & \sin 45^\circ & \cos 45^\circ \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0.707 & -0.707 \\ 0 & 0.707 & 0.707 \end{pmatrix}$$

$$R_y(30^\circ) = \begin{pmatrix} 0.866 & 0 & 0.5 \\ 0 & 1 & 0 \\ -0.5 & 0 & 0.866 \end{pmatrix}$$

Multiply $R = R_y \cdot R_x$ (column by column):

Row 1: $(0.866, 0, 0.5) \cdot$ cols of R_x : $r_{11} = 0.866$; $r_{12} = 0(0.707) + 0.5(0.707) = 0.354$; $r_{13} = 0(-0.707) + 0.5(0.707) = 0.354$

Row 2: $(0, 1, 0) \cdot$ cols: $r_{21} = 0$; $r_{22} = 0.707$; $r_{23} = -0.707$

Row 3: $(-0.5, 0, 0.866) \cdot$ cols: $r_{31} = -0.5$; $r_{32} = 0(0.707) + 0.866(0.707) = 0.612$; $r_{33} = 0(-0.707) + 0.866(0.707) = 0.612$

$$R = \begin{pmatrix} 0.866 & 0.354 & 0.354 \\ 0 & 0.707 & -0.707 \\ -0.5 & 0.612 & 0.612 \end{pmatrix}$$

(b) HTM with translation $(2, 5, 3)^T$:

$$T = \begin{pmatrix} 0.866 & 0.354 & 0.354 & 2 \\ 0 & 0.707 & -0.707 & 5 \\ -0.5 & 0.612 & 0.612 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

QB2. Given:

$$T = \begin{pmatrix} 0.707 & -0.707 & 0 & 5 \\ 0.707 & 0.707 & 0 & 3 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(a) Verify R is a valid rotation matrix: (i) unit column vectors, (ii) mutual orthogonality, (iii) $\det(R) = 1$. [4]

(b) Find T^{-1} using the property $T^{-1} = \begin{pmatrix} R^T & -R^T \vec{p} \\ \mathbf{0} & 1 \end{pmatrix}$. [3]

[7]

Solution QB2

(a) Extract R : columns $\hat{c}_1 = (0.707, 0.707, 0)^T$, $\hat{c}_2 = (-0.707, 0.707, 0)^T$, $\hat{c}_3 = (0, 0, 1)^T$.

(i) Unit vectors: $|\hat{c}_1| = \sqrt{0.5 + 0.5 + 0} = 1\checkmark$; $|\hat{c}_2| = \sqrt{0.5 + 0.5} = 1\checkmark$; $|\hat{c}_3| = 1\checkmark$

(ii) Orthogonality: $\hat{c}_1 \cdot \hat{c}_2 = (0.707)(-0.707) + (0.707)(0.707) + 0 = -0.5 + 0.5 = 0\checkmark$
 $\hat{c}_1 \cdot \hat{c}_3 = 0\checkmark$; $\hat{c}_2 \cdot \hat{c}_3 = 0\checkmark$

(iii) Determinant: Expanding along row 3: $\det(R) = 1 \cdot \det \begin{pmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{pmatrix} = 1(0.5 + 0.5) = 1\checkmark$

(b)

$$R^T = \begin{pmatrix} 0.707 & 0.707 & 0 \\ -0.707 & 0.707 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \vec{p} = \begin{pmatrix} 5 \\ 3 \\ 7 \end{pmatrix}$$

$$-R^T \vec{p} = - \begin{pmatrix} 0.707(5) + 0.707(3) \\ -0.707(5) + 0.707(3) \\ 7 \end{pmatrix} = \begin{pmatrix} -5.656 \\ 1.414 \\ -7 \end{pmatrix}$$

$$T^{-1} = \begin{pmatrix} 0.707 & 0.707 & 0 & -5.656 \\ -0.707 & 0.707 & 0 & 1.414 \\ 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

QB3. A single revolute joint has DH parameters: $\theta = 45^\circ$, $d = 0$, $a = 5$, $\alpha = 0^\circ$. Compute the 4×4 DH transformation matrix. [4]

Solution QB3

Using the standard DH matrix:

$$T = \begin{pmatrix} \cos \theta & -\sin \theta \cos \alpha & \sin \theta \sin \alpha & a \cos \theta \\ \sin \theta & \cos \theta \cos \alpha & -\cos \theta \sin \alpha & a \sin \theta \\ 0 & \sin \alpha & \cos \alpha & d \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

With $\theta = 45^\circ$, $\alpha = 0^\circ$ ($\cos \alpha = 1$, $\sin \alpha = 0$), $a = 5$, $d = 0$:

$$T = \begin{pmatrix} 0.707 & -0.707 & 0 & 3.536 \\ 0.707 & 0.707 & 0 & 3.536 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where $a \cos 45^\circ = 5(0.707) = 3.536$ and $a \sin 45^\circ = 3.536$.

QB4. A robot arm has end-effector frame:

$${}^0T_E = \begin{pmatrix} 0.866 & -0.5 & 0 & 8 \\ 0.5 & 0.866 & 0 & 4 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

A tool tip is at $(1, 0, 2)^T$ in frame $\{E\}$. Find the tool tip in the base frame $\{0\}$. [4]

Solution QB4

Use ${}^0\vec{p} = {}^0T_E \cdot {}^E\vec{p}$ with ${}^E\vec{p} = (1, 0, 2, 1)^T$:

$$\begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} 0.866 & -0.5 & 0 & 8 \\ 0.5 & 0.866 & 0 & 4 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 2 \\ 1 \end{pmatrix}$$

$$\begin{aligned}
 x &= 0.866(1) + (-0.5)(0) + 0(2) + 8 = 0.866 + 8 = 8.866 \\
 y &= 0.5(1) + 0.866(0) + 0(2) + 4 = 0.5 + 4 = 4.5 \\
 z &= 0(1) + 0(0) + 1(2) + 6 = 8
 \end{aligned}$$

$${}^0\vec{p}_{\text{tool}} = (8.866, 4.5, 8)^T$$

SECTION C — Jacobian, Singularities & Force–Torque 15 Marks

QC1. For a 2-DOF planar robot ($L_1 = L_2 = 0.5 \text{ m}$, $\theta_1 = 0^\circ$, $\theta_2 = 90^\circ$):

- Write the Jacobian $J(\theta)$ and evaluate it at the given configuration. [4]
 - Joint 1 spins at $\dot{\theta}_1 = 2 \text{ rad/s}$, joint 2 at $\dot{\theta}_2 = 1 \text{ rad/s}$. Find the end-effector velocity $(\dot{x}, \dot{y})$. [2]
 - Compute the **manipulability** $w = |\det J|$ and state what it means physically. [2]
- [8]

Solution QC1

(a) General Jacobian:

$$J = \begin{pmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{pmatrix}$$

At $\theta_1 = 0, \theta_2 = 90^\circ$: $\theta_1 + \theta_2 = 90^\circ$, so $s_1 = 0, c_1 = 1, s_{12} = 1, c_{12} = 0$.

$$J = \begin{pmatrix} -0.5(0) - 0.5(1) & -0.5(1) \\ 0.5(1) + 0.5(0) & 0.5(0) \end{pmatrix} = \begin{pmatrix} -0.5 & -0.5 \\ 0.5 & 0 \end{pmatrix}$$

(b)

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} -0.5 & -0.5 \\ 0.5 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -0.5(2) + (-0.5)(1) \\ 0.5(2) + 0 \end{pmatrix} = \begin{pmatrix} -1.5 \text{ m/s} \\ 1.0 \text{ m/s} \end{pmatrix}$$

(c)

$$\det J = (-0.5)(0) - (-0.5)(0.5) = 0 + 0.25 = 0.25$$

$$w = |\det J| = 0.25 \text{ m}^2/\text{s}$$

$w > 0$ means the robot is **not at a singularity**: it can generate end-effector velocity in any direction. Higher w = more dexterous configuration.

QC2. The Jacobian of a robot at a certain pose is:

$$J = \begin{pmatrix} -0.3 & -0.8 \\ 0.9 & 0.3 \end{pmatrix}$$

An external load applies force $\vec{F} = (20, 15)^T \text{ N}$ at the end-effector.

- Compute the required joint torques $\vec{\tau} = J^T \vec{F}$. [3]
- Verify using the virtual work principle with $\delta \vec{q} = (0.05, 0.1)^T \text{ rad}$. [4]

[7]

Solution QC2

(a)

$$J^T = \begin{pmatrix} -0.3 & 0.9 \\ -0.8 & 0.3 \end{pmatrix}$$

$$\vec{\tau} = J^T \vec{F} = \begin{pmatrix} -0.3 & 0.9 \\ -0.8 & 0.3 \end{pmatrix} \begin{pmatrix} 20 \\ 15 \end{pmatrix} = \begin{pmatrix} -6 + 13.5 \\ -16 + 4.5 \end{pmatrix} = \boxed{\begin{pmatrix} 7.5 \text{ Nm} \\ -11.5 \text{ Nm} \end{pmatrix}}$$

(b) Virtual work at joints:

$$\delta W_{\text{joints}} = \vec{\tau}^T \delta \vec{q} = 7.5(0.05) + (-11.5)(0.1) = 0.375 - 1.15 = -0.775 \text{ J}$$

Virtual end-effector displacement:

$$\delta \vec{x} = J \delta \vec{q} = \begin{pmatrix} -0.3 & -0.8 \\ 0.9 & 0.3 \end{pmatrix} \begin{pmatrix} 0.05 \\ 0.1 \end{pmatrix} = \begin{pmatrix} -0.015 - 0.08 \\ 0.045 + 0.03 \end{pmatrix} = \begin{pmatrix} -0.095 \\ 0.075 \end{pmatrix}$$

Virtual work at tip:

$$\delta W_{\text{tip}} = \vec{F}^T \delta \vec{x} = 20(-0.095) + 15(0.075) = -1.9 + 1.125 = -0.775 \text{ J} \quad \checkmark$$

Both equal -0.775 J , confirming $\vec{\tau} = J^T \vec{F}$ by the principle of virtual work.**SECTION D — DC Motors & Actuators****20 Marks**

QD1. A DC motor has a no-load speed of 60 rad/s at 120 V , with $R = 2 \Omega$ and $K = 2 \text{ Nm/A}$. It is suddenly subjected to a constant load torque of 20 Nm . Find the new steady-state speed. [5]

Solution QD1

Step 1 — Verify motor constant from no-load condition: At no-load, $I \approx 0$, so $V = K\omega_0 \implies K = V/\omega_0 = 120/60 = 2 \text{ Nm/A} \quad \checkmark$

Step 2 — Steady-state current from torque balance:

$$T_e = T_L \implies KI = T_L \implies I = \frac{T_L}{K} = \frac{20}{2} = 10 \text{ A}$$

Step 3 — New speed:

$$\omega = \frac{V - IR}{K} = \frac{120 - 10(2)}{2} = \frac{100}{2} = \boxed{50 \text{ rad/s}}$$

The load slowed the motor from 60 to 50 rad/s — a 17% speed reduction.

QD2. A robot joint motor ($V = 60 \text{ V}$, $R = 2 \Omega$, $K = 1.5 \text{ Nm/A}$) drives a load with constant friction torque 12 Nm . Bearing wear introduces an additional 3 Nm friction. Find the new steady-state speed. [5]

Solution QD2**New total load torque:** $T_L = 12 + 3 = 15 \text{ Nm}$ **Steady-state current:**

$$I = \frac{T_L}{K} = \frac{15}{1.5} = 10 \text{ A}$$

New speed:

$$\omega = \frac{V - IR}{K} = \frac{60 - 10(2)}{1.5} = \frac{40}{1.5} = \boxed{26.67 \text{ rad/s}}$$

Compare: before wear, $I = 12/1.5 = 8 \text{ A}$, $\omega_{\text{old}} = (60 - 16)/1.5 = 29.33 \text{ rad/s}$. Wear reduced speed by $\approx 9\%$.

QD3. A manipulator joint requires constant operating speed $\omega = 10 \text{ rad/s}$ against a friction load $T_L = 5 \text{ Nm}$. The motor has $R = 0.5 \Omega$, $K = 1.0 \text{ Nm/A}$. What supply voltage V must be commanded? [5]

Solution QD3**Required current:**

$$I = \frac{T_L}{K} = \frac{5}{1.0} = 5 \text{ A}$$

Required voltage (from $V = IR + K\omega$):

$$V = IR + K\omega = 5(0.5) + 1.0(10) = 2.5 + 10 = \boxed{12.5 \text{ V}}$$

This is the exact voltage the supervisory/control system must command to the power electronics.

QD4. A motor ($V = 50 \text{ V}$, $K = 1 \text{ Nm/A}$, $R = 2 \Omega$) drives a viscous load $T_L = B\omega$, currently stabilized at exactly 10 rad/s . A fluid leak reduces B by 50% . What is the new steady-state speed? [5]

Solution QD4**Step 1 — Find original B :** At steady state: $\omega_1 = \frac{VK}{BR + K^2} = 10$

$$10 = \frac{50(1)}{B(2) + 1} \implies 20B + 10 = 50 \implies B = \frac{40}{20} = 2 \text{ Nms/rad}$$

Step 2 — New $B' = 0.5B = 1 \text{ Nms/rad}$:

$$\omega_{\text{new}} = \frac{VK}{B'R + K^2} = \frac{50(1)}{1(2) + 1^2} = \frac{50}{3} = \boxed{16.67 \text{ rad/s}}$$

Lower damping means the motor speeds up to a higher equilibrium — the system is less loaded.

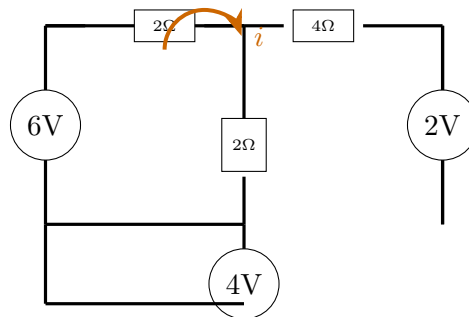
SECTION E — Circuits: Resistors, Capacitors, Inductors & Control

20 Marks

QE1. (KVL Network Analysis) Consider the circuit below: a 6 V source, 2Ω and 4Ω

in series in the top branch, a 2Ω shunt in the middle, and a 2 V source in the right branch, with a 4 V source at the bottom.

Using KVL, find the loop current i flowing clockwise in the single mesh formed by: 6 V source $\rightarrow 2\Omega \rightarrow 2\Omega$ (shunt) $\rightarrow 4\text{ V}$ source (opposing). [6]



Consider the left loop only: $+6\text{ V}$ source (rise), $-2\Omega \cdot i$ drop, $-2\Omega \cdot i$ drop across shunt, $+4\text{ V}$ source (rise at bottom). Apply KVL: [6]

Solution QE1

KVL around left mesh (clockwise current i , taking rises positive):

Starting from bottom-left, going clockwise:

- $+6\text{ V}$ (rise through battery)
- $-2i$ (drop across 2Ω)
- $-2i$ (drop across shunt 2Ω)
- -4 V (drop through 4 V bottom source, opposing direction)

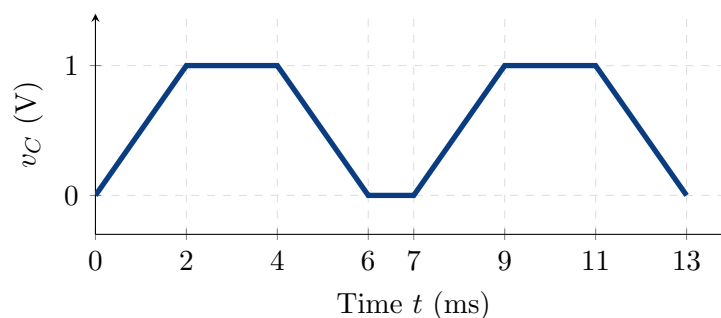
$$6 - 2i - 2i - 4 = 0$$

$$2 = 4i \implies i = 0.5\text{ A}$$

Power check:

- 6 V source delivers: $P = 6(0.5) = 3\text{ W}$ (delivered)
- 2Ω : $P = (0.5)^2(2) = 0.5\text{ W}$ (dissipated)
- Shunt 2Ω : $P = 0.5\text{ W}$ (dissipated)
- 4 V source: $P = 4(0.5) = 2\text{ W}$ (absorbed)
- Total: $3\text{ W} = 0.5 + 0.5 + 2 = 3\text{ W}$ ✓

QE2. (Capacitor Current Waveform) A capacitor $C = 2\mu\text{F}$ has the following triangular voltage waveform:



(a) Determine $i_C(t)$ for each segment of the waveform using $i_C = C dv_C/dt$. [5]

(b) Sketch $i_C(t)$ and explain the physical meaning of negative current. [2]

[7]

Solution QE2

(a) Using $i_C = C \frac{dv_C}{dt}$ with $C = 2 \mu\text{F} = 2 \times 10^{-6} \text{ F}$:

Segment	Interval	dv_C/dt	$i_C = C dv_C/dt$
Rise	$0 < t < 2 \text{ ms}$	$\frac{1-0}{2 \times 10^{-3}} = 500 \text{ V/s}$	$2 \times 10^{-6} \times 500 = +1 \text{ mA}$
Flat (high)	$2 < t < 4 \text{ ms}$	0	0 mA
Fall	$4 < t < 6 \text{ ms}$	$\frac{0-1}{2 \times 10^{-3}} = -500 \text{ V/s}$	-1 mA
Flat (low)	$6 < t < 7 \text{ ms}$	0	0 mA
Rise	$7 < t < 9 \text{ ms}$	+500 V/s	+1 mA
Flat (high)	$9 < t < 11 \text{ ms}$	0	0 mA
Fall	$11 < t < 13 \text{ ms}$	-500 V/s	-1 mA

(b) The current waveform $i_C(t)$ is a **square wave** alternating between +1 mA, 0, and -1 mA.

Physical meaning of negative current: When $dv_C/dt < 0$ (voltage falling), the capacitor is **discharging** — stored charge flows *out of* the capacitor back to the circuit. Positive current = capacitor charging (absorbing energy); negative current = discharging (releasing energy).

QE3. (RL Circuit & Motor Dynamics) A separately excited DC motor is described by $L \frac{di}{dt} + Ri + K_e \omega = V$, with $L = 10 \text{ mH}$, $R = 2 \Omega$, $K_e = K_t = 0.5$, and a step input $V = 120 \text{ V}$ at $t = 0$.

- (a) Assuming $t \ll L/R$ (neglect Ri and E_b), derive $i(t)$ and calculate i at $t = 1 \text{ ms}$. [3]
 (b) For H-bridge shoot-through on a 240 V supply with $L = 12 \mu\text{H}$ (no resistance, no back-EMF), find $\left. \frac{di}{dt} \right|_{t=0+}$ and i at $t = 1 \mu\text{s}$. [4]

[7]

Solution QE3

(a) For $t \ll L/R$, the equation reduces to $L \frac{di}{dt} \approx V$:

$$i(t) = \frac{V}{L} t = \frac{120}{10 \times 10^{-3}} t = 12000 t \text{ A}$$

At $t = 1 \text{ ms} = 10^{-3} \text{ s}$:

$$i(1 \text{ ms}) = 12000 \times 10^{-3} = \boxed{12 \text{ A}}$$

Note: $L/R = 10 \text{ mH}/2 \Omega = 5 \text{ ms}$, and $1 \text{ ms} \ll 5 \text{ ms}$, so approximation is valid here.

(b) For shoot-through, $V = L \frac{di}{dt}$ (full supply on parasitic inductance):

$$\left. \frac{di}{dt} \right|_{t=0+} = \frac{V}{L} = \frac{240}{12 \times 10^{-6}} = \boxed{20 \times 10^6 \text{ A/s} = 20 \text{ MA/s}}$$

Using linear approximation ($t = 1 \mu\text{s}$):

$$i(1 \mu\text{s}) \approx \frac{V}{L} \cdot t = 20 \times 10^6 \times 10^{-6} = \boxed{20 \text{ A}}$$

Rising from 0 to 20 A in just 1 microsecond — catastrophically destroying power devices. This is why dead-time insertion is mandatory.

LA CHEAT SHEET — Robotics & Intelligent Control

Detach this page. Only this sheet is permitted — no additional notes.

Rotation Matrices

$$R_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi & c_\phi \end{pmatrix} \quad R_y(\theta) =$$

$$\begin{pmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{pmatrix}$$

$$R_z(\psi) = \begin{pmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{YPR: } R = R_z R_y R_x$$

Properties: $R^{-1} = R^T$, $\det R = +1$, $R \in SO(3)$

Verify: cols orthonormal, $\det=1$

Homogeneous Transforms (HTM)

$$T = \begin{pmatrix} R & \vec{p} \\ \mathbf{0} & 1 \end{pmatrix} \quad T^{-1} = \begin{pmatrix} R^T & -R^T \vec{p} \\ \mathbf{0} & 1 \end{pmatrix}$$

Chain: ${}^0T_n = {}^0T_1 \cdots {}^{n-1}T_n$

Point: ${}^A\vec{p} = {}^AT_B \cdot {}^B\vec{p}$ (homogeneous: append 1)

Translate then rotate $\neq$ rotate then translate

DH Parameters & Matrix

θ_i Joint angle about Z_{i-1} (**variable, revolute**)

d_i Offset along Z_{i-1} (**variable, prismatic**)

a_i Link length along X_i (fixed)

α_i Link twist about X_i (fixed; 0 for planar)

$${}^{i-1}T_i = \begin{pmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Inverse Kinematics (2-DOF Planar)

$$\cos \theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1 L_2} \quad (\text{two solutions: } \pm)$$

$$k_1 = L_1 + L_2 \cos \theta_2, \quad k_2 = L_2 \sin \theta_2$$

$$\theta_1 = \text{atan2}(y, x) - \text{atan2}(k_2, k_1)$$

Workspace: $|L_1 - L_2| \leq r \leq L_1 + L_2$

Closed-form: exact, fast. **Numerical:** iterative (Jacobian pseudoinverse)

Jacobian & Singularities

$$\dot{\vec{x}} = J(\vec{q}) \dot{\vec{q}} \quad (\text{velocity mapping})$$

$$\text{2-DOF Planar: } J = \begin{pmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{pmatrix}$$

$$\det J = L_1 L_2 \sin \theta_2$$

Singular: $\theta_2 = 0^\circ$ (extended) or 180° (folded)

Dexterity: $w = |\det J|$ ($=0$ at singularity)

Force-torque: $\vec{\tau} = J^T \vec{F}$ (virtual work: $\vec{\tau}^T \delta \vec{q} = \vec{F}^T \delta \vec{x}$)

DC Motor Equations

Voltage: $V = IR + K_e \omega$ (steady state; $+L \frac{di}{dt}$ dynamic)

Torque: $T_e = K_t I$ ($K_e = K_t = K$ for SI units)

Back-EMF: $E_b = K_e \omega$

Steady-state speed: $\omega = \frac{V - IR}{K} = \frac{V - T_L R / K}{K}$

Stall ($\omega = 0$): $I_{\text{stall}} = V/R$, $T_{\text{max}} = KV/R$

No-load ($T_L = 0$, $I \approx 0$): $\omega_0 = V/K$

Viscous load $T_L = B\omega$: $\omega = \frac{VK}{BR + K^2}$

Const. load T_L : $I = T_L/K$, $\omega = (V - IR)/K$

Circuit Laws

Ohm: $V = IR$ **Power:** $P = VI = I^2 R = V^2/R$

KCL: $\sum I_{\text{in}} = \sum I_{\text{out}}$ at any node

KVL: $\sum V = 0$ around any closed loop

Series: $R_{\text{eq}} = \sum R_i$

Parallel: $1/R_{\text{eq}} = \sum 1/R_i$ (two: $R_{\text{eq}} = R_1 R_2 / (R_1 + R_2)$)

Capacitor & Inductor

Capacitor: $i_C = C \frac{dv_C}{dt}$ $v_C = \frac{1}{C} \int i_C dt$

DC steady state: $i_C = 0$ (open circuit)

Energy: $W_C = \frac{1}{2} C v^2$

Inductor: $v_L = L \frac{di_L}{dt}$ $i_L = \frac{1}{L} \int v_L dt$

DC steady state: $v_L = 0$ (short circuit)

Energy: $W_L = \frac{1}{2} L i^2$

Initial rate: $\left. \frac{di}{dt} \right|_{t=0^+} = \frac{V_{\text{applied}}}{L}$ (if $i(0) = 0$)

PWM, H-Bridge & MOSFET

PWM: $V_{dc} = D \cdot V_s$, $D = t_{\text{on}}/T$, $T = 1/f$

H-bridge (S1+S4 ON $\rightarrow$ forward; S2+S3 $\rightarrow$ reverse)

Shoot-through: Both same-leg switches ON $\rightarrow$ short circuit

MOSFET regions: ($V_{GS} \approx V_G$, source grounded)

OFF: $V_{GS} < V_{TH}$

Triode: $V_{DS} < V_{GS} - V_{TH}$ (low resistance, conducting)

Saturation: $V_{DS} \geq V_{GS} - V_{TH}$ (current source)

Rectifier & Trig Quick-Reference

Half-wave: $V_{dc} = V_m/\pi$, $I_{dc} = V_{dc}/R$

Full-wave: $V_{dc} = 2V_m/\pi$

θ	0°	30°	45°	$60^\circ / 90^\circ$
sin	0	0.5	0.707	0.866 / 1
cos	1	0.866	0.707	0.5 / 0

2x2 inverse: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

atan2(y, x): signed angle of vector (x, y) , range $(-\pi, \pi]$